

ALI ALATTAR

Seattle, WA | (360) 763-8538 | ali.ashraf.alattar@gmail.com | www.linkedin.com/in/ali-alattar-b059232b2 | github.com/aliashrafalattar-lab

PROFESSIONAL SUMMARY

Bioengineering and Biochemistry Honors student with a Data Science option and 2+ years of research experience across neural engineering, computational biology, protein design, electrophysiology, and experimental systems. Washington Research Foundation Fellow leading an independent project on stimulation-evoked neural responses, artifact rejection, and near real-time processing for closed-loop neuromodulation. Strong engineering/data science foundation in Python, MATLAB, Bash, signal processing, model-based analysis, hardware/software troubleshooting, documentation, and cross-functional research execution.

EDUCATION

University of Washington

Expected Aug. 2026

B.S. Bioengineering (Data Science option, Honors) & B.S. Biochemistry (Honors)

- GPA: 3.8; Washington Research Foundation Fellow; Dean's List all quarters; Engineering General Scholarship; Wayne & Grace Stanley Scholarship in Engineering.
- Relevant coursework: Neural Engineering, Biosignal Processing, Real-Time Biosignal Processing, Machine Learning, Statistics, Data Science, Computational Modeling of Bioelectricity, Bioengineering Design, Design of Experiments, Failure Analysis & Human Physiology.

South Puget Sound Community College

Jun. 2023

A.A. Biology

- GPA: 4.0; Dean's List all quarters.

RESEARCH EXPERIENCE

Neural Engineering, Repair & Design (NERD) Lab

Undergraduate Researcher / Washington Research Foundation Fellow
UW College of Engineering & UW Medicine

Apr. 2024 - Present
PI: Dr. Azadeh Yazdan-Shahmorad

Project 1: Cortico-cortical evoked potentials, artifact rejection, and closed-loop neuromodulation

- Lead an independent neural engineering project on cortico-cortical evoked potentials (CCEPs), stimulation artifact rejection, and near real-time processing for closed-loop neuromodulation.
- Designed stimulation protocols varying amplitude, pulse shape, and timing; collected and analyzed multi-channel micro-ECoG/electrophysiology recordings from rat cortex.
- Built Python pipelines for trial alignment, baseline correction, filtering, visualization, artifact-template reconstruction, subtraction, response comparison, electrode-level quality control, and near real-time analysis.
- Developed model-based and machine-learning-informed approaches, including supervised encoder-decoder modeling, to reconstruct and remove stimulation artifacts while preserving evoked neural responses.

Project 2: Synaptic plasticity, chemogenetics, and stimulation-induced cortical connectivity

- Supported a PhD-led study testing whether chemogenetic disinhibition can facilitate Hebbian-informed paired stimulation and strengthen cortical functional connectivity in rat primary somatosensory cortex.
- Veterinarian-certified for independent rodent stereotaxic brain surgery across three protocols; contributed to animal preparation, physiological monitoring, neural recording, tissue processing, and experimental validation.
- Helped with neural data filtering and interpretation, explored Kilosort4 spike sorting, and strengthened method-selection judgment by matching computational tools to the recording modality.

Institute for Protein Design, Baker Lab

Undergraduate Researcher
University of Washington

Sep. 2024 - Present
PI: Dr. David Baker

Project 1: Phosphorylation-inducible protein switches

- Worked in the IPD Jupiter cohort designing phosphorylation-inducible heterodimers / phosphotyrosine sensory domains using RFdiffusion, LigandMPNN, Rosetta/FastRelax, AlphaFold/Chai-1, Python, and Bash.
- Filtered approximately 80,000 candidate designs using RMSD, pLDDT, ipTM, PAE, contacts, clashes, hydrogen bonds, shape complementarity, interface energy, and ddG.
- Supported experimental validation through Golden Gate cloning, transformation, expression, Strep-Tactin affinity purification, size-exclusion chromatography, gel electrophoresis, purification QC, and interpretation of purified binders.

Project 2: LYTAC-based LDL degradation strategy

- Contributed to design and validation of LDL-targeting protein binders for a lysosomal degradation strategy, connecting computational binder design to disease-relevant protein clearance concepts.
- Performed recombinant protein purification and supported downstream characterization, binding assays, yeast transformation/expression workflows, yeast washing, flow-cytometry-based binding validation, and assay-data interpretation.

Project 3: BRD4-targeted protein-binder design

- Developed a protein-binder strategy targeting BRD4 bromodomains and related chromatin-reader domains, focusing on hotspot selection, binder-design strategy, candidate generation, and computational evaluation of binding feasibility.
- Used structure-guided filtering and high-throughput design organization to compare candidate quality across predicted interfaces, motif placement, steric compatibility, and cross-predictor agreement.

Project 1: Ice-cave microbial diversity and sequence interpretation

- Studied microbial diversity in ice-cave samples; performed microbial culturing, DNA purification, PCR amplification of 16S rRNA regions, gel electrophoresis verification, BLAST-based sequence interpretation, phylogenetic reasoning, and scientific report writing.

SELECTED ENGINEERING & DATA SCIENCE PROJECTS

EMG-Guided Closed-Loop Stimulation Device | BIOEN 461 Neural Engineering Tech Studio

- Designed signal-processing and control logic for an adaptive rehabilitation concept that uses EMG evidence of voluntary activation and guarding state to adjust electrical stimulation rather than deliver a fixed open-loop dose.

Ultrasonic Air Piano / Real-Time Sensing System | BIOEN 468 Real-Time Biosignal Processing

- Built Arduino-based contactless interface using ultrasonic sensors to map distance zones to real-time outputs; demonstrated embedded sensing, hardware/software integration, and live user interaction.

Computational Flow Simulation in Vascular Models | BIOEN 345 Failure Analysis & Human Physiology

- Used computational fluid simulations to analyze how vessel diameter, stenosis, and bifurcation geometry affect blood-flow dynamics, wall shear stress, and vascular-device failure considerations.

Closed-Loop Control Analysis of Insulin-GLUT4 Signaling | BIOEN 438 Biological Control Systems

- Built a control-design workflow around an SBML insulin-signaling model; identified a transfer function, analyzed root-locus stability, and proposed PI control for setpoint tracking.

Data Science and Public Health Projects | CSE Coursework

- Built regression, decision-tree, TF-IDF search, geospatial mapping, and Tableau visualization projects using Python, pandas, scikit-learn, GeoPandas, Folium, and public health datasets.

TEACHING, WORK & LEADERSHIP

Teaching Assistant - UW Department of Bioengineering

Apr. - Dec. 2025

- TA for BIOEN 316 Biosignal Processing and BIOEN 325 Biotransport/Fluid Dynamics; led sections, held office hours, graded assignments, and explained quantitative engineering concepts to diverse student groups.

Chemistry Tutor - UW Instructional Center

Apr. 2024 - Jun. 2025

- Tutored general chemistry, organic chemistry, and biochemistry for historically underserved students through individualized academic support and conceptual problem-solving guidance.

Lab Intern - Essex Laboratories LLC

Jul. 2023 - Sep. 2023

- Performed analytical chemistry quality-control testing of mint oil samples against client specifications using GC-MS, polarimetry, density measurement, and refractometry while following documented procedures in a production lab.

Undergraduate Mentorship Chair - Biomedical Engineering Society at UW

Sep. 2025 - Jun. 2026

- Led mentorship program supporting 70+ mentor-mentee pairs; managed recruitment, pairing, communication, check-ins, and program coordination.

Seattle Children's Hospital Volunteer - Inpatient Unit

Sep. 2024 - Sep. 2025

- Supported pediatric patients and families through structured play, distraction activities, one-on-one interaction, and clinical-facing communication.

PUBLICATIONS & PRESENTATIONS

- Alattar A, Shan Y, Khateeb K, Yazdan-Shahmorad A. "Robust Real-Time Artifact Processing for Brain Stimulation and Connectivity Mapping." In preparation.
- Khateeb K, Shan Y, Schwock F, Pierce A, Montalvo Vargo S, Zhou J, Stanis N, Belloir T, Alattar A, et al. "Engaging Homeostatic Mechanisms to Facilitate Stimulation-Induced Hebbian Connectivity Change." Manuscript in preparation.
- Alattar A, Shan Y, Khateeb K, Yazdan-Shahmorad A. "Robust Real-Time Artifact Processing for Brain Stimulation and Connectivity Mapping." UW Undergraduate Research Symposium, Seattle, WA, 2026.
- Alattar A, Ding K, Qing N, Demakis C, Shelley K, Baker D. "De Novo Design of Phosphotyrosine Sensory Domains." UW Undergraduate Research Symposium, 2025.

TECHNICAL SKILLS

Programming & Data: Python, MATLAB, R, Bash, Linux, Git, Jupyter, NumPy, pandas, SciPy, scikit-learn, PyTorch, matplotlib, data visualization, statistical analysis, reproducible workflows.

Engineering & Signal Processing: Biosignal processing, electrophysiology analysis, CCEP/evoked potential analysis, filtering, trial alignment, artifact rejection, supervised encoder-decoder modeling, near real-time processing, closed-loop control, hardware/software troubleshooting.

Bioengineering & Experimental: micro-ECoG/ECoG recordings, stimulation protocol design, veterinarian-certified rodent stereotaxic brain surgery, viral injection workflows, physiological monitoring, tissue processing, protein expression/purification, SEC, gel electrophoresis, binding assays, flow cytometry support.

Computational Biology & Design: RFDiffusion, AlphaFold/Chai-1, Rosetta/FastRelax, LigandMPNN, PyMOL, BLAST, structural scoring, binder-interface analysis, high-throughput filtering.

Documentation & Operations: Technical documentation, methods writing, SOP-style documentation, manuscript/poster preparation, project coordination, mentorship program management, quality-control workflows.